COLLEGE OF ENGINEERING CHENGANNUR

(Managed by IHRD, A Govt of Kerala Undertaking)
Chengannur, Alappuzha Dist. PIN: 689121

20MCA241 DATA SCIENCE LAB

LAB RECORD

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COLLEGE OF ENGINEERING CHENGANNUR

(Managed by IHRD, A Govt of Kerala Undertaking)
Chengannur, Alappuzha Dist. PIN: 689121

20MCA241 DATA SCIENCE LAB

Name: …….

Branch: ……...

Semester: ……. Roll No.: ……....

CERTIFIED BONAFIED RECORD WORK DONE BY

Reg. No.: …….

STAFF IN CHARGE

External Examiner **Internal Examiner**

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LAB CYCLE: 1 **DATE:17/07/2025**

PROGRAM NO: 1

REVIEW OF PYTHON PROGRAMMING

Aim: To review basic concepts of Python programming

1)
```
print("Demo of basic datatypes:Numbers")
x=3
y=2.5
print("x=",x)
print("y=",y)
print("Datatype of variable x:",type(x))
print("Datatype of variable y:",type(y))
print("Addition:",x+y)
print("Subtraction:",x-y)
print("Multiplication:",x*2)
print("Exponentation:",x**2)
```

Output:

```
Demo of basic datatypes:Numbers
x= 3
y= 2.5
Datatype of variable x: <class 'int'>
Datatype of variable y: <class 'float'>
Addition: 5.5
Subtraction: 0.5
Multiplication: 6
Exponentation: 9
```

2)
```
print("Demo of basic datatypes:Boolean")
t=True
f=False
print("t=",t)
print("f=",f)
print("Datatype of variable t:",type(t))
print("Datatype of variable f:",type(f))
print("Logical AND operation:",t and f)
print("Logical OR operation:",t or f)
print("Logical NOT operation:",not t)
print("Logical XOR operation:",t != f)
```

Output:

```
Demo of basic datatypes:Boolean
t= True
f= False
Datatype of variable t: <class 'bool'>
Datatype of variable f: <class 'bool'>
Logical AND operation: False
Logical OR operation: True
Logical NOT operation: False
Logical XOR operation: True
```

3)

```
print("Demo of basic datatypes:String")
s="hello"
t="world"
print("string1=",s)
print("string2=",t)
d=s+" "+t
print("String concatenation:",d)
print("Capitalize:",d.capitalize())
print("Convrted to uppercase:",s.upper())
print("Right justify a string:",s.rjust(7))
print("String at center:",s.center(7))
print("After replacing l with ell:",s.replace('l','(ell)'))
print("String after striping leading and trailing white     spaces:",'world'.strip())
```

Output:

```
Demo of basic datatypes:String
string1= hello
string2= world
String concatenation: hello world
Capitalize: Hello world
Convrted to uppercase: HELLO
Right justify a string:   hello
String at center:  hello
After replacing l with ell: he(ell)(ell)o
String after striping leading and trailing white spaces: world
```

4)

```
print("Containers:Lists")
nums=list(range(5))
print("list 'nums' contains:",nums)
nums[4]="abc"
print("List can contain elements of different types.Example:",nums)
```

```
nums.append("xyz")
print("'nums'after inserting new element at the end:",nums)
print("Sublists:")
print("A slice from index 2 to 4:",nums[2:4])
print("A slice from index 2 to the end:",nums[2:])
print("A slice from the start to index 2:",nums[:2])
print("A slice of the whole list:",nums[:])
nums[4:]=[8,9] # Assign a new sublist to a slice
print("After assign a new sublist to 'nums':")
for idx,i in enumerate(nums):
    print('%d:%s'%(idx+1,i))
even_squares=[x**2 for x in nums if x%2==0]
print("List of squares of even numbers from 'nums':",even_squares)
```

Output:

```
Containers:Lists
list 'nums' contains: [0, 1, 2, 3, 4]
List can contain elements of different types.Example: [0, 1, 2, 3, 'abc']
'nums'after inserting new element at the end: [0, 1, 2, 3, 'abc', 'xyz']
Sublists:
A slice from index 2 to 4: [2, 3]
A slice from index 2 to the end: [2, 3, 'abc', 'xyz']
A slice from the start to index 2: [0, 1]
A slice of the whole list: [0, 1, 2, 3, 'abc', 'xyz']
After assign a new sublist to 'nums':
1:0
2:1
3:2
4:3
5:8
6:9
List of squares of even numbers from 'nums': [0, 4, 64]
```

5)

```
print("Containers:Dictionaries")
d=dict()
d={'cat':'cute','dog':'furry'}
print("Dictionary:",d)
print("Is the dictionary has the key 'cat'?",'cat' in d)
d['fish']='wet'
print("After adding new entry to 'd':",d)
print("Get an element monkey:",d.get('monkey','N/A'))
print("Get an element fish:",d.get('fish','N/A'))
del d['fish']
print("After deleting the newly added entry from 'd':",d)
print("Demo of dictionary comprehension:")
squares={x:x*x for x in range(10)}
print("Squares of integers of range 10:")
```

```
for k,v in squares.items():
    print(k,":",v)
```

Output:

```
Containers:Dictionaries
Dictionary: {'cat': 'cute', 'dog': 'furry'}
Is the dictionary has the key 'cat'? True
After adding new entry to 'd': {'cat': 'cute', 'dog': 'furry', 'fish': 'wet'}
Get an element monkey: N/A
Get an element fish: wet
After deleting the newly added entry from 'd': {'cat': 'cute', 'dog': 'furry'}
Demo of dictionary comprehension:
Squares of integers of range 10:
0 : 0
1 : 1
2 : 4
3 : 9
4 : 16
5 : 25
6 : 36
7 : 49
8 : 64
9 : 81
```

6)

```
print("Containers:Sets")
num1={100,110,120}
print("Set 'num1':",num1)
num1.add(90)
print("'num1' after inserting 90:",num1)
num1.update([50,60,70])
print("'num1' after inserting multiple elements:",num1)
num1.remove(60)
print("'num1' after removing 60:",num1)
print("Set comprehension & Set operations:")
n1={x for x in range(10)}
print("n1=",n1)
n2={x for x in range(10) if x%2!=0}
print("n2=",n2)
print("n1 union n2",n1|n2)
print("n1 intersection n2",n1&n2)
print("n1 difference n2",n1-n2)
```

Output:

```
Containers:Sets
Set 'num1': {120, 100, 110}
'num1' after inserting 90: {120, 90, 100, 110}
'num1' after inserting multiple elements: {100, 70, 110, 50, 120, 90, 60}
'num1' after removing 60: {100, 70, 110, 50, 120, 90}
Set comprehension & Set operations:
n1= {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
n2= {1, 3, 5, 7, 9}
n1 union n2 {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
n1 intersection n2 {1, 3, 5, 7, 9}
n1 difference n2 {0, 2, 4, 6, 8}
```

7)

```
print("Containers:Tuples")
d={(x,x+1):x for x in range(10)}
print("Dictionary with tuple keys:")
for k,v in d.items():
    print(k,":",v)
t=(5,6)
print("Tuple t:",t)
print(d[t])
print(d[1,2])
```

Output:

```
Containers:Tuples
Dictionary with tuple keys:
(0, 1) : 0
(1, 2) : 1
(2, 3) : 2
(3, 4) : 3
(4, 5) : 4
(5, 6) : 5
(6, 7) : 6
(7, 8) : 7
(8, 9) : 8
(9, 10) : 9
Tuple t: (5, 6)
5
1
```

8)

```
print("Demo of function: Program to find factorial of a number")
def fact(n):
    if n==1:
        return 1
```

```
        else:
        return(n*fact(n-1))
n=int(input("Enter a number:"))
print("Factorial:",fact(n))
```

Output:

```
Demo of function: Program to find factorial of a number
Factorial: 120
```

9)

```
class Greeter:
    def __init__(self,name):
        self.name=name
    def greet(self,loud=False):
        if loud:
            print('HELLO, %s!'%self.name.upper())
        else:
            print('Hello, %s'%self.name)
g=Greeter('Fred')
g.greet()
g.greet(loud=True)
```

Output:

```
Hello, Fred
HELLO, FRED!
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:29/07/2025**

PROGRAM NO: 2

TRANSPOSE OF MATRIX

Aim: To write a program to find Transpose of a matrix

Program:

```
import numpy as np
rows = int(input("Enter the number of rows: "))
cols = int(input("Enter the number of columns: "))
matrix = []
print("Enter the elements row by row: ")
for i in range (rows):
    row=[]
    for j in range (cols):
        row.append(int(input()))
    matrix.append(row)
print("matrix 1:\n ")
for i in range(rows):
    for j in range(cols):
        print(matrix[i][j], end=" ")
    print()

print("Transpose:\n")
matrixa=np.array(matrix)
trans1=matrixa.T
print(trans1)
```

Output:

```
Enter the number of rows:  2
Enter the number of columns:  2
Enter the elements row by row:
 1
 2
 3
 4
matrix 1:

1 2
3 4
Transpose:

[[1 3]
 [2 4]]
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:29/07/2025**

PROGRAM NO: 3

RANK OF MATRIX

Aim: To write a program to find Rank of a matrix

Program:

```
import numpy as np
rows = int(input("Enter the number of rows: "))
cols = int(input("Enter the number of columns: "))
matrix = []
print("Enter the elements row by row: ")
for i in range (rows):
    row=[]
    for j in range (cols):
        row.append(int(input()))
    matrix.append(row)
print("2D matrix:\n ")
for i in range(rows):
    for j in range(cols):
        print(matrix[i][j], end=" ")
    print()
print("Rank of the matrix: \n")
matrixa=np.array(matrix)
rank_matrix=np.linalg.matrix_rank(matrixa)
print(rank_matrix)
```

Output:

Enter the number of rows: 2

Enter the number of columns: 2

Enter the elements row by row:

2

4

6

8

2D matrix:

2 4

6 8

Rank of the matrix:

2

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:29/07/2025**

PROGRAM NO: 4

DOT PRODUCT OF MATRIX

Aim: To write a program to find Dot Product of a matrix

Program:

```
import numpy as np
a=int(input("Enter no.of rows:"))
b=int(input("Enter no.of columns:"))
matrixa= []

print("Enter elements row wise:")
for i in range(a):
    c=[]
    for j in range(b):
        c.append(int(input()))
    matrixa.append(c)

d=int(input("Enter no.of rows:"))
e=int(input("Enter no.of columns:"))
matrixb= []

if b==d:
    print("Enter elements row wise:")
    for i in range(d):
        f=[]
        for j in range(e):
            f.append(int(input()))
```

```
        matrixb.append(f)
    result=np.dot(matrixa,matrixb)
    print("Dot product:")
    print(result)
else:
    print("Error! Row and column size should be same")
```

Output:

```
Enter no.of rows: 2
Enter no.of columns: 2
Enter elements row wise:
 1
 2
 3
 4
Enter no.of rows: 2
Enter no.of columns: 2
Enter elements row wise:
 5
 6
 7
 8
Dot product:
[[19 22]
 [43 50]]
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:07/08/2025**

PROGRAM NO: 5

INVERSE OF MATRIX

Aim: To write a program to find Inverse of a matrix

Program:

```
import numpy as np
rows = int(input("Enter the number of rows: "))
cols = int(input("Enter the number of columns: "))
matrix = []
print("Enter the elements row by row: ")
for i in range (rows):
    row=[]
    for j in range (cols):
        row.append(int(input()))
    matrix.append(row)
print("2D matrix:\n ")
for i in range(rows):
    for j in range(cols):
        print(matrix[i][j], end=" ")
    print()
print("Inverse Matrix: \n")
matrixa=np.array(matrix)
inv_matrix=np.linalg.inv(matrixa)
print(inv_matrix
```

Output:

```
Enter the number of rows:  2
Enter the number of columns:  2
Enter the elements row by row:
 1
 2
 3
 4
2D matrix:

1 2
3 4
Inverse Matrix:

[[-2.   1. ]
 [ 1.5 -0.5]]
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:07/08/2025**

PROGRAM NO: 6

SVD OF MATRIX

Aim: To write a program to find SVD of a matrix

Program:

```
import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6],
              [7, 8, 9]])
U, s, Vt = np.linalg.svd(A)
print("Matrix U (Left singular vectors):\n", U)
print("\nSingulpyplot as ar values (as a 1D array):\n", s)
print("\nMatrix Vt (Transpose of Right singular vectors):\n", Vt)
Sigma = np.zeros((A.shape[0], A.shape[1]))
Sigma[:A.shape[1], :A.shape[1]] = np.diag(s)
A_reconstructed = U @ Sigma @ Vt
print("\nReconstructed matrix:\n", A_reconstructed)
```

Output:

Matrix U (Left singular vectors):

[[-0.21483724 0.88723069 0.40824829]

[-0.52058739 0.24964395 -0.81649658]

[-0.82633754 -0.38794278 0.40824829]]

Singulpyplot as ar values (as a 1D array):

[1.68481034e+01 1.06836951e+00 4.41842475e-16]

Matrix Vt (Transpose of Right singular vectors):

[[-0.47967118 -0.57236779 -0.66506441]

[-0.77669099 -0.07568647 0.62531805]

[-0.40824829 0.81649658 -0.40824829]]

Reconstructed matrix:

[[1. 2. 3.]

[4. 5. 6.]

[7. 8. 9.]]

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:07/08/2025**

PROGRAM NO: 7

DATA VISUALIZATION

Aim: To write a program using matplotlib to analyse the data on graph using bar chart, boxplot, bubble chart, line chart, histogram, scatterplot, pie chart.

Program:

```
import matplotlib.pyplot as plt
#BARCHART
categories=['A','B','C']
values=[10,25,15]
plt.bar(categories,values)
plt.title("BarChartExample")
plt.xlabel("Categories")
plt.ylabel("Values")
plt.show()

#BOXPLOT
data = [ [10, 12, 14, 15, 18, 20, 22],
        [8, 9, 11, 13, 17, 19, 21],
        [14, 16, 18, 20, 23, 25, 27] ]
plt.boxplot(data)
plt.title("BoxplotExample")
plt.xlabel("Groups")
plt.ylabel("Values")
plt.show()

#BUBBLECHART
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 1, 5, 3]
sizes = [100, 300, 50, 400, 200] # Represents the third variable
plt.scatter(x, y, s=sizes, alpha=0.5)
plt.title('Bubble Chart Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()

#LINE CHART
X=[10,20,30,40]
Y=[20,25,35,55]
plt.plot(X,Y)
plt.title("LineplotExample")
plt.xlabel("X-Axis")
plt.ylabel("Y-axis")
plt.show()

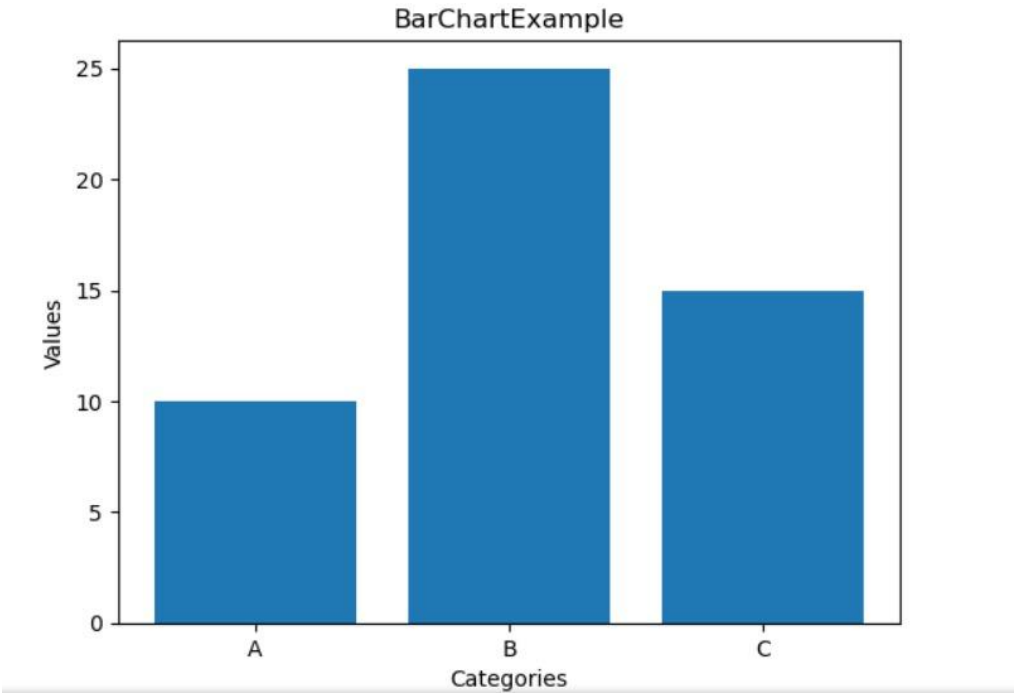
#HISTOGRAM
data = [1, 2, 2, 3, 3, 3, 4, 4, 5]
plt.hist(data, bins=5)
plt.title('Histogram Example')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.show()

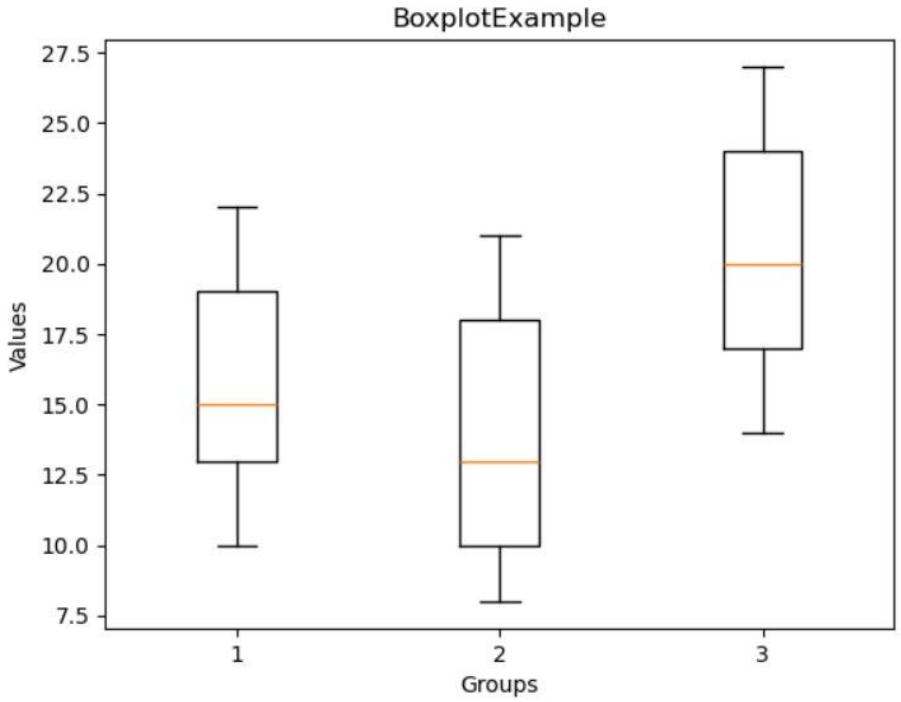
#SCATTER PLOT
x = [1, 2, 3, 4, 5]
y = [2, 4, 1, 5, 3]
plt.scatter(x, y)
plt.title('Scatter Plot Example')
```

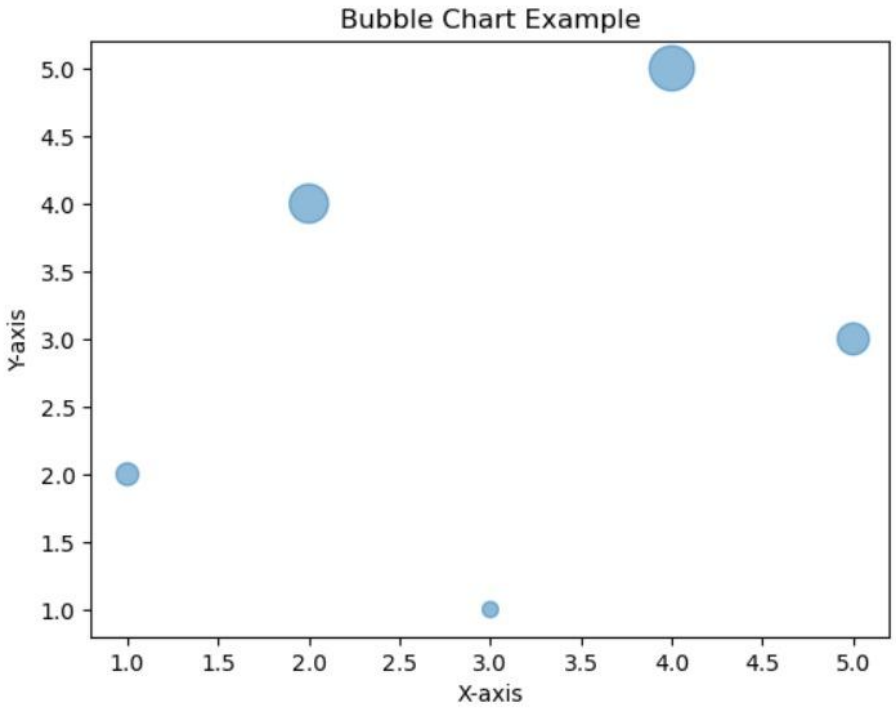
```
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()

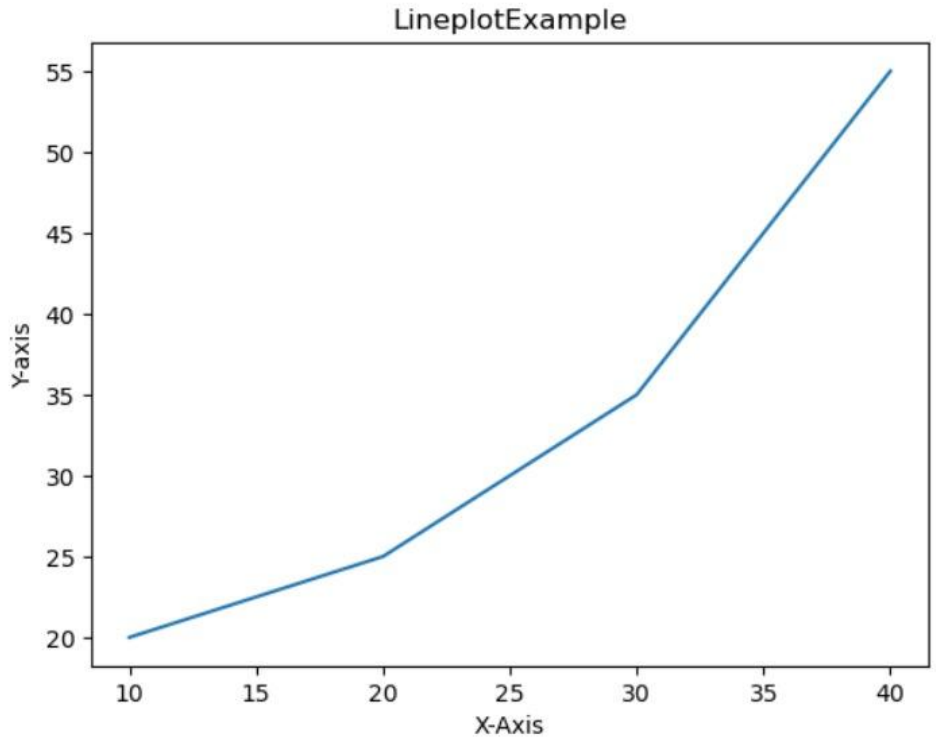
#PIE CHART
labels = ['Category A', 'Category B', 'Category C']
sizes = [30, 45, 25]
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90)
plt.title('Pie Chart Example')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```

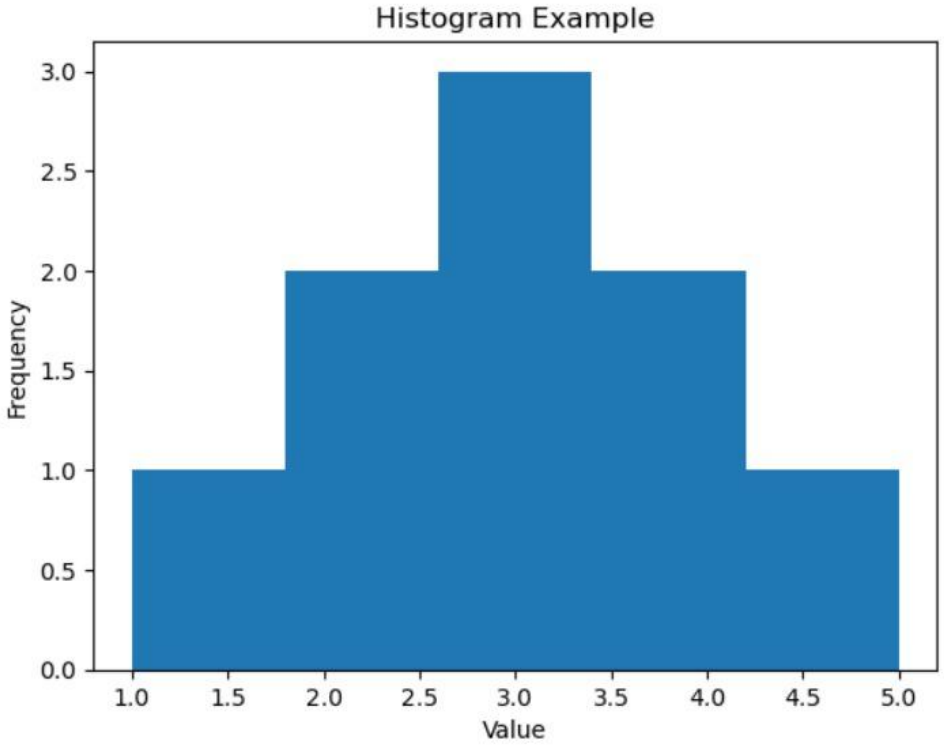
Output:

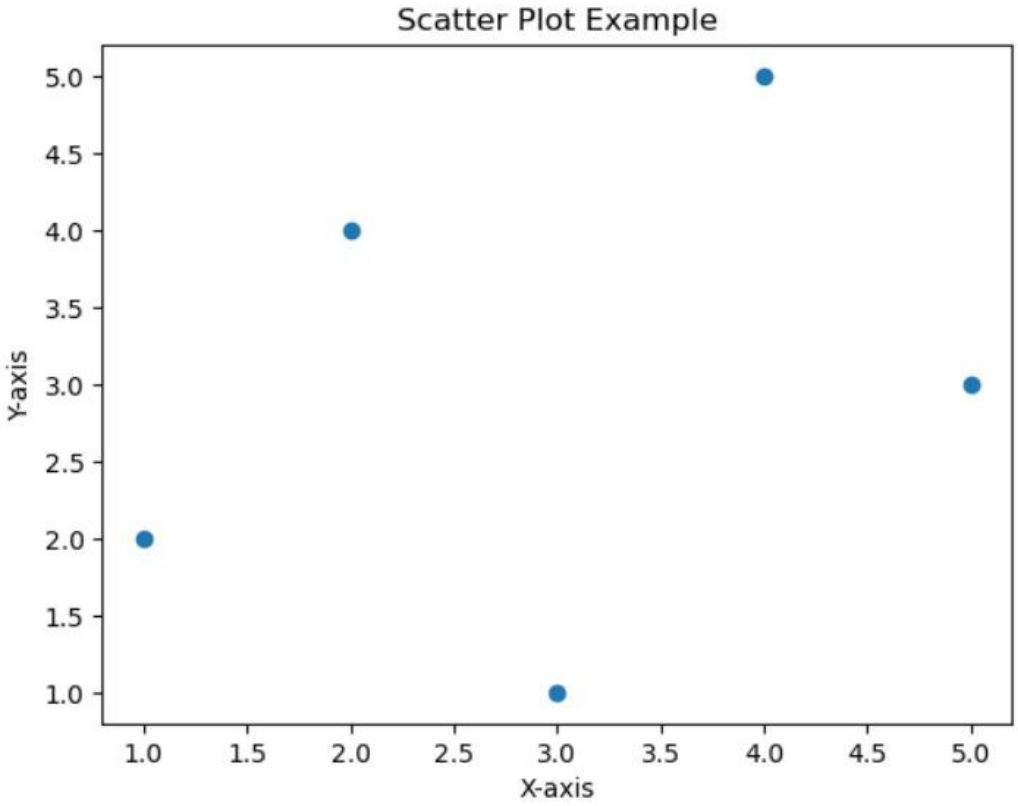


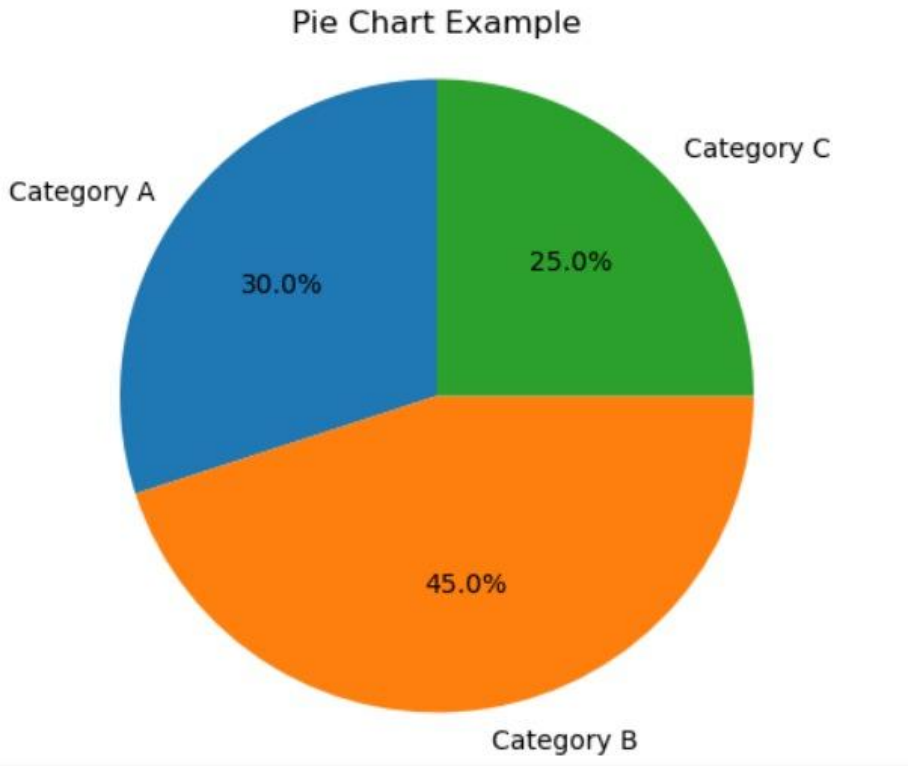

BoxplotExample
Values
Groups
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
1
2
3


Bubble Chart Example
Y-axis
X-axis
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


LineplotExample
Y-axis
X-Axis
55
50
45
40
35
30
25
20
10
15
20
25
30
35
40


Histogram Example
Frequency
Value
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0





Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:14/08/2025**

PROGRAM NO: 8

DATA HANDLING USING PANDAS

Aim: To create a dataset for student data analysis (having fields name, department, marks and grade) and perform the following operations:

a) Load the dataset and display first five records, also display the total number of students.

```
import pandas as pd

data={"Name":["Abhi","Raju","Bob","Alice","John","Sam"],
      "Department":["CSE","ECE","IT","CSE","IT","ECE"],
      "Marks":[80,70,90,60,77,None]}
df=pd.DataFrame(data)
df.head()

total_students=len(df)
print("Total number of students:",total_students)
```

Output:

	Name	Department	Marks
0	Abhi	CSE	80
1	Raju	ECE	70
2	Bob	IT	90
3	Alice	CSE	60
4	John	IT	77

Total number of students: 6

b) Display the count of students in each department

```
print("Count of students in each department")
b=df.groupby('Department').size()
print(b)
```

Output:

```
Count of students in each department
Department
CSE    2
ECE    2
IT     2
dtype: int64
```

c) Find the students who scored more than 75 marks

```
print("Students with mark greater than 75:")
stud=df[df['Marks']>75]
stud
```

Output:

Students with mark greater than 75:

	Name	Department	Marks
0	Abhi	CSE	80
2	Bob	IT	90
4	John	IT	77

d) Fill any missing mark with average mark of the class

```
avgm=df['Marks'].mean()
df['Marks'].fillna(avgm, inplace=True)
df
```

Output:

	Name	Department	Marks
0	Abhi	CSE	80.0
1	Raju	ECE	70.0
2	Bob	IT	90.0
3	Alice	CSE	60.0
4	John	IT	77.0
5	Sam	ECE	75.4

e) Add a new column "Grade" based on the marks

```
def assign_grade(marks):
    if marks >= 90:
        return 'A'
    elif marks >= 80:
        return 'B'
    elif marks >= 70:
        return 'C'
    elif marks >= 60:
        return 'D'
    else:
        return 'F'

df['Grade']=df['Marks'].apply(assign_grade)
df
```

Output:

	Name	Department	Marks	Grade
0	Abhi	CSE	80.0	B
1	Raju	ECE	70.0	C
2	Bob	IT	90.0	A
3	Alice	CSE	60.0	D
4	John	IT	77.0	C
5	Sam	ECE	75.4	C

f) Display department wise average marks

```
print("Department wise average score:")
avg_dept=df.groupby('Department')['Marks'].mean()
avg_dept
```

Output:

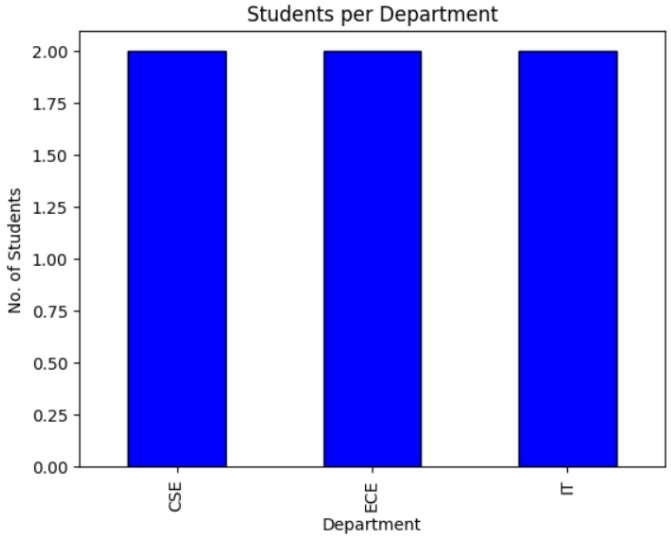
```
Department wise average score:
Out[14]:
Department
CSE    70.0
ECE    72.7
IT     83.5
Name: Marks, dtype: float64
```

g) Plot a bar chart showing the number of student per department

```
import matplotlib.pyplot as plt
dept_count = df['Department'].value_counts()
dept_count.plot(kind="bar", color="blue", edgecolor="black")

plt.title("Students per Department")
plt.xlabel("Department")
plt.ylabel("No. of Students")
plt.show()
```

Output:

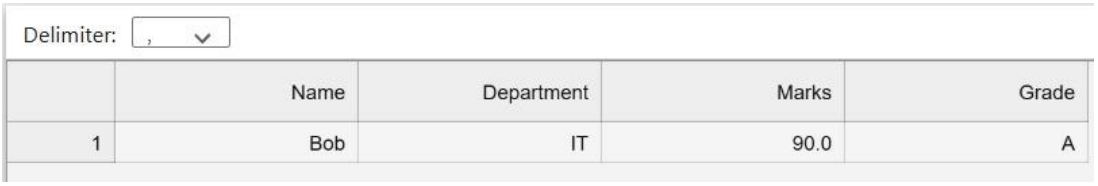


h) Export the list of students with marks greater than or equal to 90 into a new csv file

```
high_scores=df[df['Marks'] >=90]
high_scores.to_csv("high_scores.csv",index=False)
print("CSV file created successfully")
```

Output:

CSV file created successfully


Delimiter: ,

	Name	Department	Marks	Grade
1	Bob	IT	90.0	A

Result: The program was executed successfully and output verified.

LAB CYCLE: 1 **DATE:11/09/2025**

EXPERIMENT NO: 9

DATA HANDLING USING PANDAS

Aim: To read a dataset of Chipotle order into a data frame and perform the following operations:

a) Display the first 5 rows of the dataframe

```
import pandas as pd
df=pd.read_csv('chipotle.tsv', sep="\t")
df.head()
```

Output:

	order_id	quantity	item_name	choice_description	item_price
0	1	1	Chips and Fresh Tomato Salsa	NaN	$2.39
1	1	1	Izze	[Clementine]	$3.39
2	1	1	Nantucket Nectar	[Apple]	$3.39
3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	$2.39
4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans...	$16.98

b) Sort the data frame based on the item price column in descending order and display the first 5 rows

```
df['item_price']=df['item_price'].str.replace('$',"",regex=False).astype(float)

df_sorted=df.sort_values(by="item_price",ascending=False)
print(df_sorted.head())
```

Output:

```
      order_id  quantity                     item_name  \
3598      1443        15  Chips and Fresh Tomato Salsa
3480      1398         3                 Carnitas Bowl
3602      1443         4               Chicken Burrito
1254       511         4               Chicken Burrito
3601      1443         3                Veggie Burrito

                                choice_description  item_price
3598                                           NaN       44.25
3480  [Roasted Chili Corn Salsa, [Fajita Vegetables,...       35.25
3602  [Fresh Tomato Salsa, [Rice, Black Beans, Chees...       35.00
1254  [Fresh Tomato Salsa, [Fajita Vegetables, Rice,...       35.00
3601  [Fresh Tomato Salsa, [Fajita Vegetables, Rice,...       33.75
```

c) Find and display records where item_name contain the substring “Chicken”

```
print(df[df['item_name'].str.contains("chicken",case=False)])
```

Output:

```
      order_id  quantity             item_name  \
4            2         2          Chicken Bowl
5            3         1          Chicken Bowl
11           6         1  Chicken Crispy Tacos
12           6         1    Chicken Soft Tacos
13           7         1          Chicken Bowl
...        ...       ...                   ...
4604      1828         1          Chicken Bowl
4615      1832         1    Chicken Soft Tacos
4619      1834         1    Chicken Salad Bowl
4620      1834         1    Chicken Salad Bowl
4621      1834         1    Chicken Salad Bowl

                                choice_description  item_price
4     [Tomatillo-Red Chili Salsa (Hot), [Black Beans...       16.98
5     [Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou...       10.98
11    [Roasted Chili Corn Salsa, [Fajita Vegetables,...        8.75
12    [Roasted Chili Corn Salsa, [Rice, Black Beans,...        8.75
13    [Fresh Tomato Salsa, [Fajita Vegetables, Rice,...       11.25
...                                                 ...         ...
4604  [Fresh Tomato Salsa, [Rice, Black Beans, Chees...        8.75
4615   [Fresh Tomato Salsa, [Rice, Cheese, Sour Cream]]        8.75
4619  [Fresh Tomato Salsa, [Fajita Vegetables, Pinto...       11.25
4620  [Fresh Tomato Salsa, [Fajita Vegetables, Lettu...        8.75
4621  [Fresh Tomato Salsa, [Fajita Vegetables, Pinto...        8.75

[1560 rows x 5 columns]
```

d) Remove square bracket from the choice_description column using negex

```
df['choice_description']=df['choice_description'].str.replace(r"[\[\]]","",regex=True)
df.head()
```

Output:

	order_id	quantity	item_name	choice_description	item_price
0	1	1	Chips and Fresh Tomato Salsa	NaN	2.39
1	1	1	Izze	Clementine	3.39
2	1	1	Nantucket Nectar	Apple	3.39
3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	2.39
4	2	2	Chicken Bowl	Tomatillo-Red Chili Salsa (Hot), Black Beans, ...	16.98

e) Display the datatypes of each column in the dataframe

```
print(df.dtypes)
```

Output:

```
order_id               int64
quantity               int64
item_name             object
choice_description    object
item_price           float64
dtype: object
```

f) Replace '$' in item_price column and convert the column to float_type

```
df['item_price']=df['item_price'].str.replace('$',"",regex=False).astype(float)

print(df['item_price'].dtype)
```

Output:

float64

Result: The program was executed successfully and output verified.

LAB CYCLE: 2 **DATE:18/09/2025**

PROGRAM NO: 10

K-Nearest Neighbor

Aim: To implement K-Nearest Neighbor algorithm using any dataset available in public domain and evaluate its accuracy.

Program:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn import metrics
iris=load_iris()
x=iris.data
y=iris.target
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
c_knn=KNeighborsClassifier(n_neighbors=3)
c_knn.fit(x_train,y_train)
y_pred=c_knn.predict(x_test)
print("Accuracy:",metrics.accuracy_score(y_test,y_pred))
sample=[[2,2,3,3]]
pred=c_knn.predict(sample)
pred_v=[str(iris.target_names[p]) for p in pred]
print(pred_v)
```

Output:

```
Accuracy: 0.9777777777777777
['versicolor']
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 2 **DATE:18/09/2025**

PROGRAM NO: 11

Naive Bayes

Aim: To implement Naive Bayes classification algorithm on the following data of a certain set of patients seen by a doctor and to check whether the doctor can conclude that a person having chills, fever, mild headache and without running nose has flu?

Chills	Running nose	Headache	Fever	Has flu
Y	N	mild	Y	N
Y	Y	no	N	Y
Y	N	strong	Y	Y
N	Y	mild	Y	Y
N	N	no	N	N
N	Y	strong	N	Y
N	Y	strong	N	N
Y	Y	mild	Y	Y

Program:

```
from sklearn.naive_bayes import CategoricalNB
from sklearn import preprocessing
# Data
chills = ['Y', 'Y', 'Y', 'N', 'N', 'N', 'N', 'Y']
running_nose = ['N', 'Y', 'N', 'Y', 'N', 'Y', 'Y', 'Y']
headache = ['mild', 'no', 'strong', 'mild', 'no', 'strong', 'strong', 'mild']
fever = ['Y', 'N', 'Y', 'Y', 'N', 'Y', 'N', 'Y']
has_flu = ['N', 'Y', 'Y', 'Y', 'N', 'Y', 'N', 'Y']
# Encode and train
le = preprocessing.LabelEncoder()
features = list(zip(le.fit_transform(chills),
```

```
le.fit_transform(running_nose),
le.fit_transform(headache),
                le.fit_transform(fever)))
label = le.fit_transform(has_flu)
model = CategoricalNB()
model.fit(features, label)
# Predict
predicted = model.predict([[1, 0, 0, 1]])  # Y, N, mild, Y
print("Prediction:", le.inverse_transform(predicted)[0])
print("Accuracy:", model.score(features, label))
```

Output:

```
Prediction: Y
Accuracy: 0.75
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 2 **DATE:25/09/2025**

PROGRAM NO: 12

Decision Tree

Aim: To implement Decision Tree classification algorithm using any dataset available in public

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

# Load the dataset
data = load_iris()
print("Data shape:", data.data.shape)
print("Classes to predict:", data.target_names)
print("Features:", data.feature_names)

# Define features and target
X = data.data
y = data.target
print("Feature shape:", X.shape)
print("Target shape:", y.shape)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=50, test_size=0.25)
```

```
# -------------------- Model 1: Gini --------------------
clf_gini = DecisionTreeClassifier()
clf_gini.fit(X_train, y_train)
y_pred_gini = clf_gini.predict(X_test)

print("\n--- Using Gini ---")
print("Accuracy on train data:", accuracy_score(y_train, clf_gini.predict(X_train)))
print("Accuracy on test data:", accuracy_score(y_test, y_pred_gini))

# -------------------- Model 2: Entropy --------------------
clf_entropy = DecisionTreeClassifier(criterion='entropy')
clf_entropy.fit(X_train, y_train)
y_pred_entropy = clf_entropy.predict(X_test)

print("\n--- Using Entropy ---")
print("Accuracy on train data:", accuracy_score(y_train, clf_entropy.predict(X_train)))
print("Accuracy on test data:", accuracy_score(y_test, y_pred_entropy))

# -------------------- Model 3: Entropy + min_samples_split --------------------
clf_entropy_split = DecisionTreeClassifier(criterion='entropy', min_samples_split=50)
clf_entropy_split.fit(X_train, y_train)
y_pred_entropy_split = clf_entropy_split.predict(X_test)

print("\n--- Using Entropy with min_samples_split=50 ---")
print("Accuracy on train data:", accuracy_score(y_train, clf_entropy_split.predict(X_train)))
print("Accuracy on test data:", accuracy_score(y_test, y_pred_entropy_split))

# -------------------- Visualization of the tree --------------------
plt.figure(figsize=(12, 8))
plot_tree(clf_entropy,
```

```
        filled=True,
        feature_names=data.feature_names,
        class_names=data.target_names)
plt.title("Decision Tree (Entropy)")
plt.show()
```

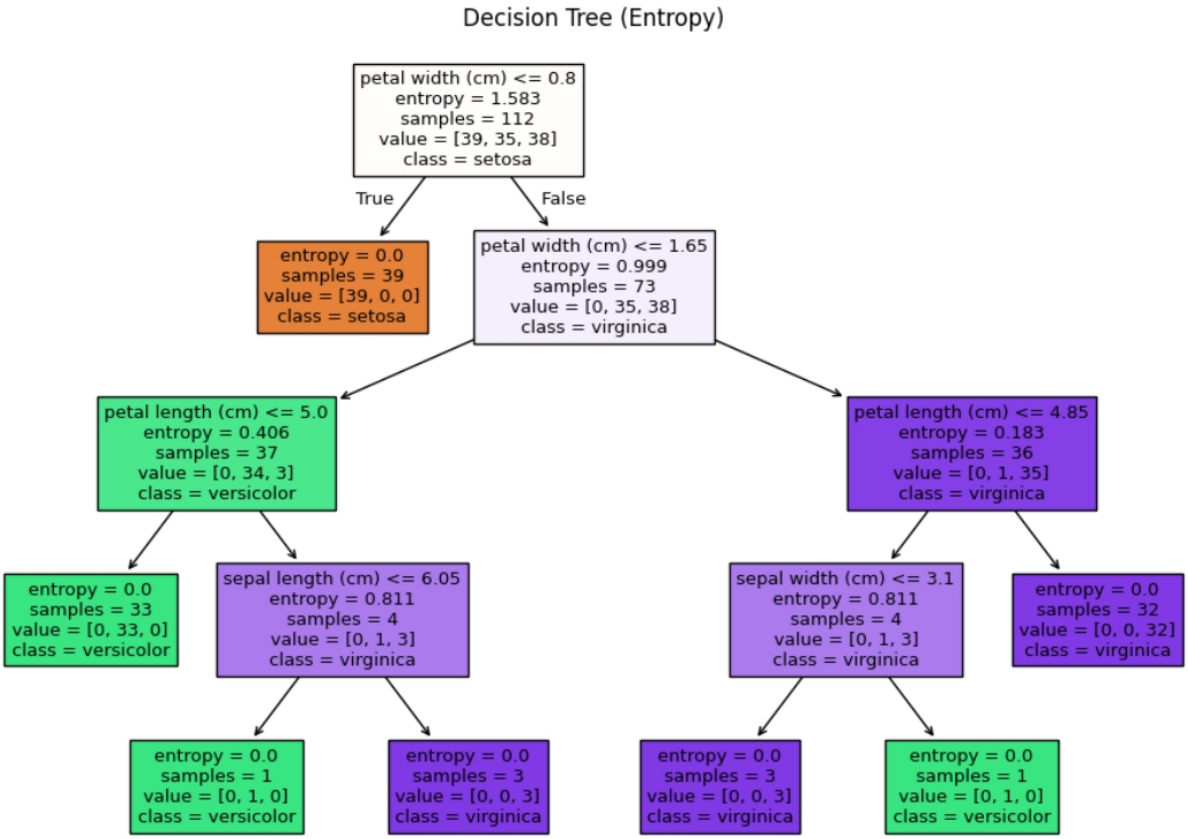
Output:

```
Data shape: (150, 4)
Classes to predict: ['setosa' 'versicolor' 'virginica']
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Feature shape: (150, 4)
Target shape: (150,)

--- Using Gini ---
Accuracy on train data: 1.0
Accuracy on test data: 0.9473684210526315

--- Using Entropy ---
Accuracy on train data: 1.0
Accuracy on test data: 0.9473684210526315

--- Using Entropy with min_samples_split=50 ---
Accuracy on train data: 0.9642857142857143
Accuracy on test data: 0.9473684210526315
```



Result: The program was executed successfully and output verified.

LAB CYCLE: 2 **DATE:09/10/2025**

PROGRAM NO: 13

Linear Regression

Aim: To implement linear regression techniques using standard dataset available in public

Program:

```
#Load sklearn Libraries
#import libraries
import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets,linear_model
from sklearn.metrics import mean_squared_error, r2_score

#Load Data
#load diabetes dataset

diabetesX,diabetesY=datasets.load_diabetes(return_X_y=True)

#split datasets
diabetesX = diabetesX[:,np.newaxis,2]

#split the data into training/testing sets
diabetesXtrain=diabetesX[:-20]
diabetesXtest=diabetesX[-20:]

#split the targets into training/testing sets
diabetesYtrain=diabetesY[:-20]
diabetesYtest=diabetesY[-20:]
```

```
#Creating the model
#create linear regression object
regr=linear_model.LinearRegression()

#train the model using training sets
regr.fit(diabetesXtrain,diabetesYtrain)

#Make prediction
#make predictions using the testing set
diabetesYpred=regr.predict(diabetesXtest)

#finding coefficients and mean square error
#coefficients
print("Coefficients:\n",regr.coef_)

#mean squared error
print("Mean squared error:\n",mean_squared_error(diabetesYtest,diabetesYpred))

#coefficient of determination:1 is perfect prediction
print("Coefficient of determination:",r2_score(diabetesYtest,diabetesYpred))
plt.scatter(diabetesXtest,diabetesYtest,color='black')
plt.plot(diabetesXtest,diabetesYpred,color='blue',linewidth=3)
plt.xticks()
plt.yticks()
plt.show()
```

Output:

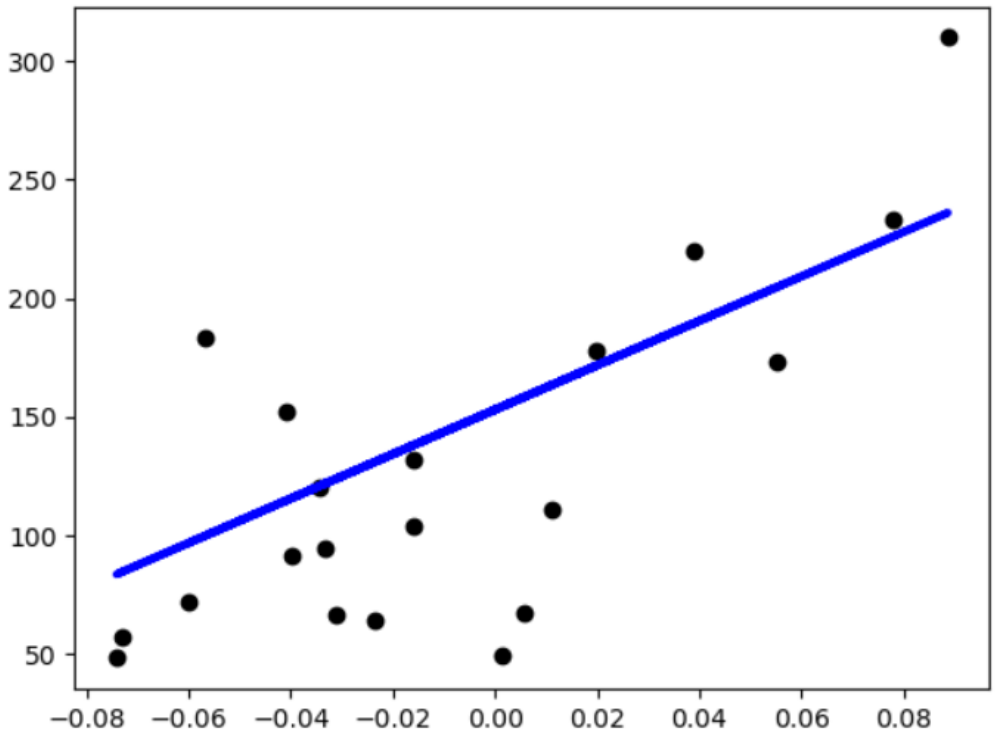
Coefficients:

[938.23786125]

Mean squared error:

2548.07239872597

Coefficient of determination: 0.47257544798227147



Result: The program was executed successfully and output verified.

LAB CYCLE: 2 **DATE:09/10/2025**

PROGRAM NO: 14

Support Vector Machine

Aim: To implement Support Vector Machine using standard dataset available in public and evaluate its accuracy.

Program:

```
from sklearn.model_selection import train_test_split
from sklearn import datasets
from sklearn import svm
from sklearn import metrics
cancer=datasets.load_breast_cancer()
x_train, x_test, y_train, y_test=train_test_split(cancer.data,cancer.target,test_size=0.3,
random_state=109)
clf=svm.SVC(kernel='linear')
clf.fit(x_train, y_train)
y_pred=clf.predict(x_test)
print("Actual values:", y_test)
print("Predicted values:", y_pred)
print("Accuracy:", metrics.accuracy_score(y_test, y_pred))
print("Precision:", metrics.precision_score(y_test, y_pred))
print("Recall:", metrics.recall_score(y_test, y_pred))
```

Output:

```
Actual values: [1 1 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 1 0 1 1 0 0 1 1 0 1 1 1 1 0 1 1 1 1 1
 0 1 1 0 1 0 1 1 1 0 1 0 0 1 1 0 1 1 0 1 1 1 0 0 1 0 1 0 0 1 1 1 1 0 1 1 1
 0 1 0 1 1 1 1 1 1 1 0 1 1 1 1 1 0 0 1 0 1 1 1 1 1 1 0 0 1 1 0 1 1 0 1 0 1 1
 0 1 1 0 0 0 1 0 1 1 1 1 1 0 1 0 1 0 1 0 0 0 0 0 0 1 1 0 1 1 1 0 1 1 0 0 1 0
 0 0 1 1 1 1 1 0 0 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1]
Predicted values: [1 1 0 0 1 0 1 1 1 0 0 0 1 0 0 1 0 0 1 0 1 1 0 0 1 1 0 1 1 1 1 0 1 1 1 1 1
 0 1 1 0 1 0 1 1 1 1 1 0 0 1 1 0 1 1 0 1 1 1 0 0 1 0 1 0 0 1 1 1 1 0 1 1 1
 0 1 0 1 1 1 1 1 1 1 0 0 1 1 1 1 0 0 0 0 1 1 1 1 1 0 0 1 0 0 1 1 0 1 0 1 1
 0 1 1 0 0 0 1 0 1 1 1 1 1 0 1 0 1 0 1 0 0 0 1 0 1 1 0 1 1 1 0 1 1 0 0 1 0
 0 0 1 1 1 1 1 0 0 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1]
Accuracy: 0.9649122807017544
Precision: 0.9811320754716981
Recall: 0.9629629629629629
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 3 **DATE:23/10/2025**

PROGRAM NO: 15

Convolutional Neural Network

Aim: To perform image classification using Convolutional Neural Networks.

Program:

```
#------CNN------
# FOR MODULE ERROR; ADD NEW CELL -> TYPE AND RUN:-  pip install tensorflow

import tensorflow as tf
from tensorflow import keras
from keras import datasets, layers, models
import matplotlib.pyplot as plt

# Load and normalize the CIFAR-10 dataset
(train_images, train_labels), (test_images, test_labels) = datasets.cifar10.load_data()
train_images, test_images = train_images / 255.0, test_images / 255.0

# Class names for CIFAR-10
class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']

# Plotting the first 25 images from the training set
plt.figure(figsize=(10, 10))
for i in range(25):
    plt.subplot(5, 5, i + 1)
    plt.xticks([])
    plt.yticks([])
    plt.grid(False)



    plt.imshow(train_images[i])
```

```
    plt.xlabel(class_names[train_labels[i][0]])
plt.show()

# Building the CNN model
model = models.Sequential()
model.add(layers.Conv2D(32, (3, 3), activation='relu',
input_shape=(32, 32, 3)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))

# Adding Dense layers
model.add(layers.Flatten())
model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10))
print("Architecture of the model:\n")
model.summary()

# Compiling the model
model.compile(optimizer='adam',
loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True) , metrics=['accuracy'])

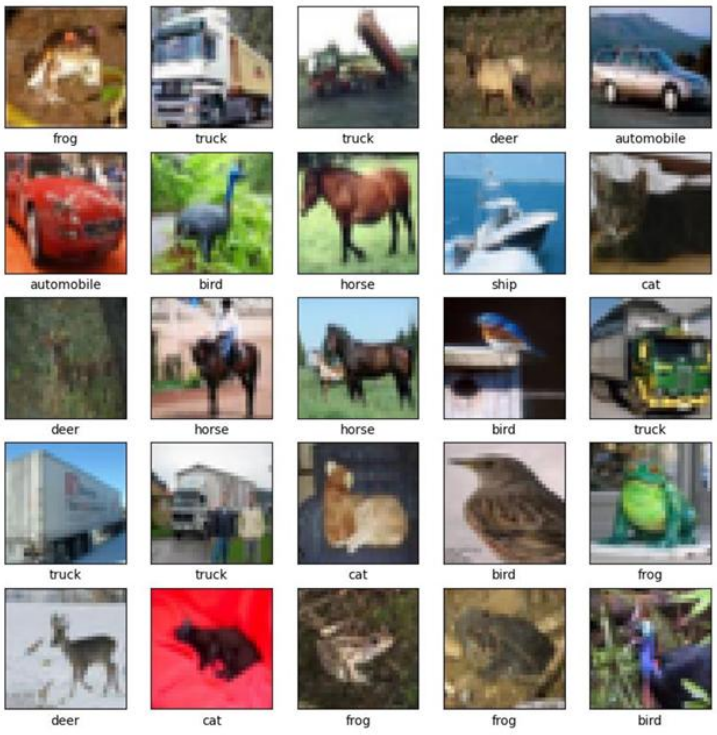
# Training the model
history = model.fit(train_images, train_labels, epochs=10, validation_data=(test_images,
test_labels))
# Evaluating the model
test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)



print("Test loss:", test_loss)
print("Test accuracy:", test_acc)
```

```
# Plotting training and validation accuracy/loss
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label='val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(loc='lower right')
plt.title('Training and Validation Accuracy')
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='loss')
plt.plot(history.history['val_loss'], label='val_loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend(loc='upper right')
plt.title('Training and Validation Loss')
plt.show()
```

Output:



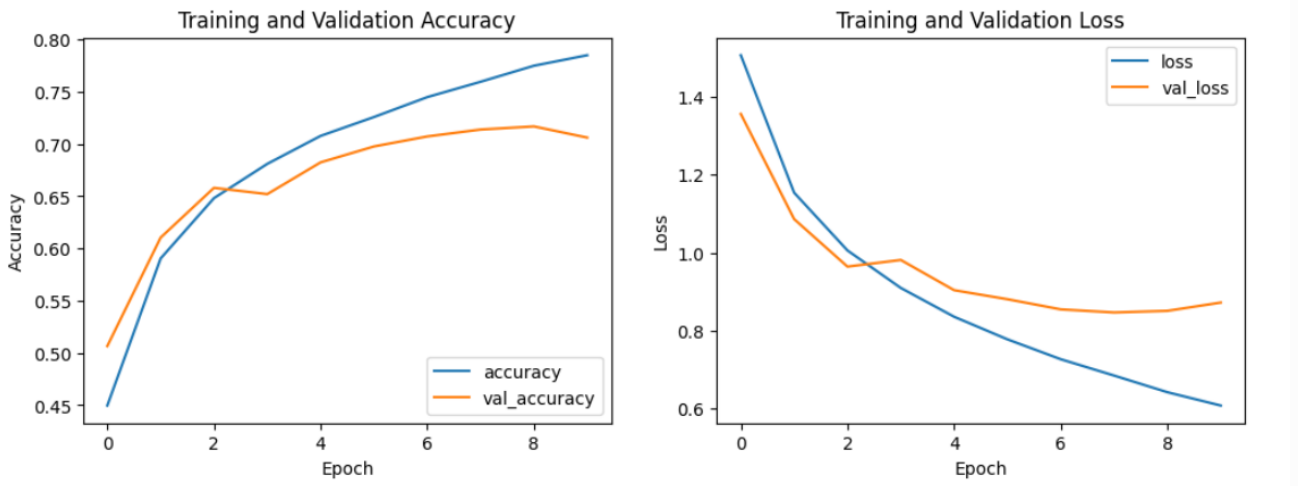
Architecture of the model:

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 64)	36,928
flatten (Flatten)	(None, 1024)	0
dense (Dense)	(None, 64)	65,600
dense_1 (Dense)	(None, 10)	650

Total params: 122,570 (478.79 KB)
Trainable params: 122,570 (478.79 KB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 86s 48ms/step - accuracy: 0.3603 - loss: 1.7316 - val_accuracy: 0.5066 - val_loss: 1.3563
Epoch 2/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 74s 48ms/step - accuracy: 0.5698 - loss: 1.2132 - val_accuracy: 0.6104 - val_loss: 1.0862
Epoch 3/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 81s 52ms/step - accuracy: 0.6429 - loss: 1.0161 - val_accuracy: 0.6580 - val_loss: 0.9642
Epoch 4/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 94s 60ms/step - accuracy: 0.6779 - loss: 0.9157 - val_accuracy: 0.6520 - val_loss: 0.9813
Epoch 5/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 103s 66ms/step - accuracy: 0.7067 - loss: 0.8387 - val_accuracy: 0.6824 - val_loss: 0.9037
Epoch 6/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 73s 46ms/step - accuracy: 0.7280 - loss: 0.7715 - val_accuracy: 0.6976 - val_loss: 0.8805
Epoch 7/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 88s 56ms/step - accuracy: 0.7459 - loss: 0.7196 - val_accuracy: 0.7072 - val_loss: 0.8544
Epoch 8/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 58s 37ms/step - accuracy: 0.7627 - loss: 0.6751 - val_accuracy: 0.7138 - val_loss: 0.8465
Epoch 9/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 96s 46ms/step - accuracy: 0.7781 - loss: 0.6282 - val_accuracy: 0.7168 - val_loss: 0.8507
Epoch 10/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 82s 52ms/step - accuracy: 0.7931 - loss: 0.5843 - val_accuracy: 0.7061 - val_loss: 0.8719
313/313 - 6s - 19ms/step - accuracy: 0.7061 - loss: 0.8719
Test loss: 0.8718963861465454
Test accuracy: 0.7060999870300293
```



Result: The program was executed successfully and output verified.

LAB CYCLE: 3 **DATE:23/10/2025**

PROGRAM NO: 16

NLTK

Aim: To implement Part of Speech tagging, N-gram, Smoothening and Chunking using NLTK.

Program:

```
import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('averaged_perceptron_tagger')
nltk.download('maxent_ne_chunker')

nltk.download('punkt_tab')

from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk import RegexpParser

# Sample text
text = """A newspaper is the strongest medium for news. People are
reading newspapers for decades.
It has a huge contribution to globalization. Right now because of
easy internet connection,
people don't read printed newspapers often. They read the online
version."""
```

```
print("Sample text: \n", text, "\n")

# Tokenizing by sentence
sent_tokenized = sent_tokenize(text)

print("Tokenizing by sentence: \n", sent_tokenized, "\n")

# Tokenizing by word
word_tokenized = word_tokenize(text)
print("Tokenizing by word: \n", word_tokenized, "\n")

# Removing stop words and punctuation
stop_words = set(stopwords.words('english'))
punctuation_set = set(punctuation)

print("After filtering the stop words and punctuation: ")
filtered_words = [word for word in word_tokenized if word.casefold()
not in stop_words and word.casefold() not in punctuation_set]
for word in filtered_words:
    print(word)

# Stemming
ps = PorterStemmer()
words = ["reading", "globalization", "Being", "Went", "gone",
"going"]
print("\nGiven words: ", words)
stemm = [ps.stem(i) for i in words]
print("After stemming: ", stemm, "\n")

# Lemmatization
```

```
lem = WordNetLemmatizer()
print("rocks:", lem.lemmatize("rocks"))
print("corpora:", lem.lemmatize("corpora"))
print("better:", lem.lemmatize("better"))
print("believes:", lem.lemmatize("believes"), "\n")

# Lemmatization with POS tag
print("went as adjective:", lem.lemmatize("went", pos="a"))
print("went as verb:", lem.lemmatize("went", pos="v"))
print("went as noun:", lem.lemmatize("went", pos="n"), "\n")

# POS tagging
nltk.download('averaged_perceptron_tagger_eng')
postag = nltk.pos_tag(word_tokenized)
print("POS tagging: \n")
for i in postag:
    print(i)
print("\n")

# Chunking
grammar = "NP: {<DT>?<JJ>*<NN>}"
chunker = RegexpParser(grammar)
output = chunker.parse(postag)
print("After Chunking:\n", output)
output.pretty_print()
```

Output:

```
Sample text:
 A newspaper is the strongest medium for news. People are
reading newspapers for decades.
It has a huge contribution to globalization. Right now because of
easy internet connection,
people don't read printed newspapers often. They read the online
version.

Tokenizing by sentence:
 ['A newspaper is the strongest medium for news.', 'People are \nreading newspapers for decades.', 'It has a huge contributio
n to globalization.', "Right now because of \neasy internet connection,  \npeople don't read printed newspapers often.", 'The
y read the online \nversion.']

Tokenizing by word:
 ['A', 'newspaper', 'is', 'the', 'strongest', 'medium', 'for', 'news', '.', 'People', 'are', 'reading', 'newspapers', 'for',
'decades', '.', 'It', 'has', 'a', 'huge', 'contribution', 'to', 'globalization', '.', 'Right', 'now', 'because', 'of', 'eas
y', 'internet', 'connection', ',', 'people', 'do', "n't", 'read', 'printed', 'newspapers', 'often', '.', 'They', 'read', 'th
e', 'online', 'version', '.']

After filtering the stop words and punctuation:
newspaper
strongest
medium
news
People
reading
newspapers
decades
huge
contribution
globalization
Right
easy
internet
connection
people
n't
read
printed
newspapers
often
read
online
version

Given words:  ['reading', 'globalization', 'Being', 'Went', 'gone', 'going']
After stemming:  ['read', 'global', 'be', 'went', 'gone', 'go']

rocks: rock
corpora: corpus
better: better
believes: belief

went as adjective: went
went as verb: go
went as noun: went

POS tagging:

('A', 'DT')
('newspaper', 'NN')
('is', 'VBZ')
('the', 'DT')
('strongest', 'JJS')
('medium', 'NN')
('for', 'IN')
('news', 'NN')
('.', '.')
('People', 'NNS')
('are', 'VBP')
('reading', 'VBG')
('newspapers', 'NNS')
('for', 'IN')
('decades', 'NNS')
('.', '.')
('It', 'PRP')
('has', 'VBZ')
('a', 'DT')
('huge', 'JJ')
('contribution', 'NN')
('to', 'TO')
('globalization', 'NN')
('.', '.')
('Right', 'RB')
('now', 'RB')
('because', 'IN')
('of', 'IN')
('easy', 'JJ')
('internet', 'JJ')
('connection', 'NN')
(',', ',')
('people', 'NNS')
('do', 'VBP')
("n't", 'RB')
('read', 'VB')
('printed', 'JJ')
('newspapers', 'NNS')
('often', 'RB')
('.', '.')
('They', 'PRP')
('read', 'VBD')
('the', 'DT')
('online', 'JJ')
('version', 'NN')
('.', '.')
```

```
After Chunking:
 (S
  (NP A/DT newspaper/NN)
  is/VBZ
  the/DT
  strongest/JJS
  (NP medium/NN)
  for/IN
  (NP news/NN)
  ./.
  People/NNS
  are/VBP
  reading/VBG
  newspapers/NNS
  for/IN
  decades/NNS
  ./.
  It/PRP
  has/VBZ
  (NP a/DT huge/JJ contribution/NN)
  to/TO
  (NP globalization/NN)
  ./.
  Right/RB
  now/RB
  because/IN
  of/IN
  (NP easy/JJ internet/JJ connection/NN)
  ,/,
  people/NNS
  do/VBP
  n't/RB
  read/VB
  printed/JJ
  newspapers/NNS
  often/RB
  ./.
  They/PRP
  read/VBD
  (NP the/DT online/JJ version/NN)
  ./.)

S
   ____________________________________________________________________________________________________|_____________________________________________________________________________
_____________________________________________________________________________________________________|_____________________________
_____________________________________________________________________________________________________|_____________________________
_____________________
  |      |          |         |     |     |         |         |           |         |         |      |    |      |
|    |     |       |        |      |     |     |        |      |       |         |          |     |     |
|      |       NP                NP       NP          NP                        NP                    NP
NP
  |      |         |         |     |     |         |         |           |         |         |      |    |      |
|    |     |       |        |      |     |     |        |      |       |         |          |     |     |
|      |    ____|______           |       |     _____|__________             |           ________|___________
________|________
is/VBZ the/DT strongest/JJS for/IN ./. People/NNS are/VBP reading/VBG newspapers/NNS for/IN decades/NNS ./. It/PRP has/VBZ t
o/TO ./. Right/RB now/RB because/IN of/IN ,/, people/NNS do/VBP n't/RB read/VB printed/JJ newspapers/NNS often/RB ./. They/PR
P read/VBD ./. A/DT     newspaper/NN medium/NN news/NN a/DT huge/JJ contribution/NN globalization/NN easy/JJ internet/JJ conn
ection/NN the/DT online/JJ version/NN
```

Result: The program was executed successfully and output verified.

LAB CYCLE: 3 **DATE:23/10/2025**

PROGRAM NO: 17

Web-Crawler and Scrapping Web Pages

Aim: To implement a simple web-crawler and scrapping web pages.

Program:

```
import requests
import numpy as np
import pandas as pd
from bs4 import BeautifulSoup
import matplotlib.pyplot as plt
import re
import os
%matplotlib inline
riturl = "https://purdue.edu"
webpage = requests.get(riturl)
ritsoup = BeautifulSoup(webpage.content, "html.parser")
print(ritsoup)
print("Title of the parsed page: ", ritsoup.title.string)
print()
print("All the links: ")
links = [link.get('href') for link in ritsoup.find_all('a')]
print(links, "\n")
print(ritsoup.head)
print("For over 150 years, generations of Boilermakers have left their mark in small steps and giant leaps. Today, we continue in those footsteps as we bring our best and learn to build a better world, together.")
```

```
print(ritsoup.head.meta)
```

```
paragraphs = ritsoup.find_all("p")


print()
print("Get all <p> elements: ", paragraphs)
print()
print("Gets all the <p> elements with a class attribute with value 'hide': ",        ritsoup.find_all("p",
attrs={"class": "hide"}))
print()
if ritsoup.h1:
    print("Obtaining strings: ", ritsoup.h1.string)
print()
if ritsoup.h1:
    print("Obtaining strings using contents method: ",
ritsoup.h1.contents)
print()
if len(paragraphs) > 2:
    print(paragraphs[2], "\n")
    print(paragraphs[2].string, "\n")
if len(paragraphs) > 2:
    para2 = paragraphs[2].contents
    print(para2, "\n")
if len(paragraphs) > 5:
    para7 = paragraphs[5].contents
    print(para7, "\n")
if len(paragraphs) > 3:
    print("Raw text from the 4th paragraph:")
    for s in paragraphs[3].stripped_strings:
        print("=" * 50)
        print(s)
```

Output:

```
<!DOCTYPE html>

<html class="is-fullheight" lang="en-US">
<head>
<title>Purdue University</title>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1" name="viewport"/>
<link href="http://gmpg.org/xfn/11" rel="profile"/>
<style type="text/css">@media (min-width:1024px){
.purdue-home-news-events__list .purdue-home-news-events__title{
min-height:3.5rem;
}
}
.purdue-home-hot-spot {
    overflow: visible !important;
}
.purdue-home-slide__hot-spot-mobile{
    overflow: hidden;
}

/***
.purdue-home-button-list li:last-child .purdue-home-button{
    background: #000;
    color: #fff;
}
.purdue-home-button-list li:last-child .purdue-home-button:after{
    background-image: url("data:image/svg+xml,%3Csvg xmlns=%27http://www.w3.org/2000/svg%27 viewBox=%270 0 448 512%27%3E%3C%21--%21 Font Awesome Pro 6.4.0 by @fontawesome - https://fontawesome.com License - https://fontawesome.com/license %28Commercial License%29 Copyright 2023 Fonticons, Inc. --%3E%3Cpath fill=%27%23cfb991%27 d=%27M438.6 278.6c12.5-12.5 12.5-32.8 0-45.3l-160-160c-12.5-12.5-32.8-12.5-45.3 0s-12.5 32.8 0 45.3L338.8 224 32 224c-17.7 0-32 14.3-32 32s14.3 32 32 32l306.7 0L233.4 393.4c-12.5 12.5-12.5 32.8 0 45.3s32.8 12.5 45.3 0l160-160z%27/%3E%3C/svg%3E");
}
.purdue-home-button-list li:last-child .purdue-home-button:hover{
color: #000;
background: #cfb991 !important;
}
.purdue-home-button-list li:last-child .purdue-home-button:hover:after{
background-image: url("data:image/svg+xml,%3Csvg xmlns=%27http://www.w3.org/2000/svg%27 viewBox=%270 0 448 512%27%3E%3C%21--%21 Font Awesome Pro 6.4.0 by @fontawesome - https://fontawesome.com License - https://fontawesome.com/license %28Commercial License%29 Copyright 2023 Fonticons, Inc. --%3E%3Cpath fill=%27%2300000%27 d=%27M438.6 278.6c12.5-12.5 12.5-32.8 0-45.3l-160-160c-12.5-12.5-32.8-12.5-45.3 0s-12.5 32.8 0 45.3L338.8 224 32 224c-17.7 0-32 14.3-32 32s14.3 32 32 32l306.7 0L233.4 393.4c-12.5 12.5-12.5 32.8 0 45.3s32.8 12.5 45.3 0l160-160z%27/%3E%3C/svg%3E");
}
***/</style> <style>img:is([sizes="auto" i], [sizes^="auto," i]) { contain-intrinsic-size: 3000px 1500px }</style>
<!-- Google Tag Manager for WordPress by gtm4wp.com -->
<script data-cfasync="false" data-pagespeed-no-defer="">
        var gtm4wp_datalayer_name = "dataLayer";
        var dataLayer = dataLayer || [];
</script>
<!-- End Google Tag Manager for WordPress by gtm4wp.com -->
<!-- The SEO Framework by Sybre Waaijer -->
<meta content="max-snippet:-1,max-image-preview:large,max-video-preview:-1" name="robots">
<link href="https://www.purdue.edu/home/" rel="canonical">
<meta content="Purdue University is a world-renowned, public research university that advances discoveries in science, technology, engineering and math." name="description">
<meta content="website" property="og:type"/>
<meta content="en_US" property="og:locale"/>
<meta content="Purdue University" property="og:site_name"/>
<meta content="Purdue University" property="og:title"/>
<meta content="Purdue University is a world-renowned, public research university that advances discoveries in science, technology, engineering and math." property="og:description"/>
<meta content="https://www.purdue.edu/home/" property="og:url"/>
<meta content="https://www.purdue.edu/home/wp-content/uploads/2023/09/cropped-Home-Cover-2023_KAL_1419.jpg" property="og:image"/>
<meta content="1499" property="og:image:width"/>
<meta content="787" property="og:image:height"/>
d math.","inLanguage":"en-US","isPartOf":{"@id":"https://www.purdue.edu/home/#/schema/WebSite"},"breadcrumb":{"@type":"BreadcrumbList","@id":"https://www.purdue.edu/home/#/schema/BreadcrumbList","itemListElement":{"@type":"ListItem","position":1,"name":"Purdue University"}},"potentialAction":{"@type":"ReadAction","target":"https://www.purdue.edu/home/"},"about":{"@id":"https://www.purdue.edu/home/#/schema/Organization"}},{"@type":"Organization","@id":"https://www.purdue.edu/home/#/schema/Organization","name":"Purdue University","url":"https://www.purdue.edu/home/"}]}</script>
<!-- / The SEO Framework by Sybre Waaijer | 24.91ms meta | 10.87ms boot -->
<link href="//use.typekit.net" rel="dns-prefetch"/>
<link href="//fonts.googleapis.com" rel="dns-prefetch"/>
<link href="//use.fontawesome.com" rel="dns-prefetch"/>
<script type="text/javascript">
/* <![CDATA[ */
window._wpemojiSettings = {"baseUrl":"https:\/\/s.w.org\/images\/core\/emoji\/16.0.1\/72x72\/","ext":".png","svgUrl":"https:\/\/s.w.org\/images\/core\/emoji\/16.0.1\/svg\/","svgExt":".svg","source":{"concatemoji":"https:\/\/www.purdue.edu\/home\/wp-includes\/js\/wp-emoji-release.min.js?ver=6.8.2"}};
/*! This file is auto-generated */
!function(s,n){var o,i,e;function c(e){try{var t={supportTests:e,timestamp:(new Date).valueOf()};sessionStorage.setItem(o,JSON.stringify(t))}catch(e){}}function p(e,t,n){e.clearRect(0,0,e.canvas.width,e.canvas.height),e.fillText(t,0,0);var t=new Uint32Array(e.getImageData(0,0,e.canvas.width,e.canvas.height).data),a=(e.clearRect(0,0,e.canvas.width,e.canvas.height),e.fillText(n,0,0),new Uint32Array(e.getImageData(0,0,e.canvas.width,e.canvas.height).data));return t.every(function(e,t){return e===a[t]})}function u(e,t){e.clearRect(0,0,e.canvas.width,e.canvas.height),e.fillText(t,0,0);for(var n=e.getImageData(16,16,1,1),a=0;a<n.data.length;a++)if(0!==n.data[a])return!1;return!0}function f(e,t,n,a){switch(t){case"flag":return n(e,"\ud83c\udff3\ufe0f\u200d\u26a7\ufe0f","\ud83c\udff3\ufe0f\u200b\u26a7\ufe0f")?!1:!n(e,"\ud83c\udde8\ud83c\uddf6","\ud83c\udde8\u200b\ud83c\uddf6")&&!n(e,"\ud83c\udff4\udb40\udc67\udb40\udc62\udb40\udc65\udb40\udc6e\udb40\udc67\udb40\udc7f","\ud83c\udff4\u200b\udb40\udc67\u200b\udb40\udc62\u200b\udb40\udc65\u200b\udb40\udc6e\u200b\udb40\udc67\u200b\udb40\udc7f");case"emoji":return!a(e,"\ud83e\uded4")}return!1}function g(e,t,n,a){var r="undefined"!=typeof WorkerGlobalScope&&self instanceof WorkerGlobalScope?new OffscreenCanvas(300,150):s.createElement("canvas"),o=r.getContext("2d",{willReadFrequently:!0}),i=(o.textBaseline="top",o.font="600 32px Arial",{});return e.forEach(function(e){i[e]=t(o,e,n,a)}),i}function t(e){var t=s.createElement("script");t.src=e,t.defer=!0,s.head.appendChild(t)}"undefined"!=typeof Promise&&(o="wpEmojiSettingsSupports",i=["flag","emoji"],n.supports={everything:!0,everythingExceptFlag:!0},e=new Promise(function(e){s.addEventListener("DOMContentLoaded",e,{once:!0})}),new Promise(function(t){var n=function(){try{var e=JSON.parse(sessionStorage.getItem(o));if("object"==typeof e&&"number"==typeof e.timestamp&&(new Date).valueOf()<e.timestamp+604800&&"object"==typeof e.supportTests)return e.supportTests}catch(e){}return null}();if(!n){if("undefined"!=typeof Worker&&"undefined"!=typeof OffscreenCanvas&&"undefined"!=typeof URL&&URL.createObjectURL&&"undefined"!=typeof Blob)try{var e="postMessage("+g.toString()+"("+[JSON.stringify(i),f.toString(),p.toString(),u.toString()].join(",")+"));",a=new Blob([e],{type:"text/javascript"}),r=new Worker(URL.createObjectURL(a),{name:"wpTestEmojiSupports"});return void(r.onmessage=function(e){c(n=e.data),r.terminate(),t(n)})}catch(e){}c(n=g(i,f,p,u))}t(n)}).then(function(e){for(var t in e)n.supports[t]=e[t],n.supports.everything=n.supports.everything&&n.supports[t],"flag"!==t&&(n.supports.everythingExceptFlag=n.supports.everythingExceptFlag&&n.supports[t]);n.supports.everythingExceptFlag=n.supports.everythingExceptFlag&&!n.supports.flag,n.DOMReady=!1,n.readyCallback=function(){n.DOMReady=!0}}).then(function(){return e}).then(function(){var e;n.supports.everything||(n.readyCallback(),(e=n.source||{}).concatemoji?t(e.concatemoji):e.wpemoji&&e.twemoji&&(t(e.twemoji),t(e.wpemoji)))}))((window,document),window._wpemojiSettings);
/* ]]> */
</script>
<style id="wp-emoji-styles-inline-css" type="text/css">

        img.wp-smiley, img.emoji {
                display: inline !important;
                border: none !important;
                box-shadow: none !important;
                height: 1em !important;
                width: 1em !important;
                margin: 0 0.07em !important;
                vertical-align: -0.1em !important;
                background: none !important;
                padding: 0 !important;
        }
```

```
        }
</style>
<link href="https://www.purdue.edu/home/wp-includes/css/dist/block-library/common.min.css?ver=6.8.2" id="wp-block-library-cs
s" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-includes/css/classic-themes.min.css?ver=6.8.2" id="classic-theme-styles-css" media
="all" rel="stylesheet" type="text/css"/>
<link href="https://use.typekit.net/ghc8hdz.css?ver=6.8.2" id="brandfonts-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/unitedsans.css?ver=6.8.2" id="unitedsans-css" media
="all" rel="stylesheet" type="text/css"/>
<link href="https://fonts.googleapis.com/css2?family=Source+Serif+Pro%3Awght%40400%3B600%3B700&amp;display=swap&amp;ver=6.8.
2" id="sourceserif-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/plugins/wp-accessibility/css/wpa-style.css?ver=2.2.0" id="wpa-style-css" m
edia="all" rel="stylesheet" type="text/css"/>
<style id="wpa-style-inline-css" type="text/css">
:root { --admin-bar-top : 7px; }
</style>
<link href="https://use.fontawesome.com/releases/v6.4.2/css/all.css?ver=6.8.2" id="load-fa-css" media="all" rel="stylesheet"
type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/style.css?ver=6.8.2" id="purdueBrand-style-css" m
edia="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/app.css?ver=64131b13d3f628a0dd7f" id="purdu
eBrand-brand-style-css" media="" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/blocks/hot-spot/style-index.css?ver=64131b1
3d3f628a0dd7f" id="purdue-hot-spot-style-css-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/blocks/mixed-rtb/style-index.css?ver=64131b
13d3f628a0dd7f" id="purdue-mixed-rtb-style-css-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/blocks/news-and-events/style-index.css?ver=
64131b13d3f628a0dd7f" id="purdue-news-and-events-style-css-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/blocks/search-block/style-index.css?ver=641
31b13d3f628a0dd7f" id="purdue-search-block-style-css-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-content/themes/purdue-home-theme/build/blocks/canopy-hero/style-index.css?ver=6413
1b13d3f628a0dd7f" id="purdue-canopy-hero-style-css-css" media="all" rel="stylesheet" type="text/css"/>
<link href="https://www.purdue.edu/home/wp-json/" rel="https://api.w.org/"/><link href="https://www.purdue.edu/home/wp-json/w
p/v2/pages/12" rel="alternate" title="JSON" type="application/json"/><link href="https://www.purdue.edu/home/xmlrpc.php?rsd"
rel="EditURI" title="RSD" type="application/rsd+xml"/>
<link href="https://www.purdue.edu/home/wp-json/oembed/1.0/embed?url=https%3A%2F%2Fwww.purdue.edu%2Fhome%2F" rel="alternate"
title="oEmbed (JSON)" type="application/json+oembed"/>
<link href="https://www.purdue.edu/home/wp-json/oembed/1.0/embed?url=https%3A%2F%2Fwww.purdue.edu%2Fhome%2F&amp;format=xml" r
el="alternate" title="oEmbed (XML)" type="text/xml+oembed"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/favicon.ico" rel="shortcut icon" type="imag
e/x-icon"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/apple-icon-57x57.png" rel="apple-touch-ico
n" sizes="57x57"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/apple-icon-60x60.png" rel="apple-touch-ico
n" sizes="60x60"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/apple-icon-72x72.png" rel="apple-touch-ico
n" sizes="72x72"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/apple-icon-76x76.png" rel="apple-touch-ico
n" sizes="76x76"/>
<link href="https://www.purdue.edu/home/wp-content/mu-plugins/boilerup-wp/favicon/apple-icon-114x114.png" rel="apple-touch-ic
on" sizes="114x114"/>
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<p class="purdue-home-news-events__title">Record-setting expenditures reflect university’s national leadership in critical se
ctors</p>
<p class="purdue-home-news-events__date">
<span class="date">October 10, 2025</span>
</p>
</a>
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-card__content">Dive into future-focused fields like AI, biomedical engineering, data science, cybersecurity, sports technolo
gy and more.  </p>, <p class="cta-link">Explore majors</p>, <p class="purdue-home-cta-card__tag">Purdue University in Indiana
polis</p>, <p class="purdue-home-cta-card__title">Career-ready students</p>, <p class="purdue-home-cta-card__content">Prepare
for the jobs of tomorrow through hands-on learning opportunities, internships and co-ops.  </p>, <p class="cta-link">Learn Mo
re</p>, <p class="purdue-home-cta-card__tag">Purdue Memorial Union</p>, <p class="purdue-home-cta-card__title">Built for Boil
ermakers</p>, <p class="purdue-home-cta-card__content">Meet at the union for food and retail shops, plus the Union Club Hote
l.  </p>, <p class="cta-link">Learn More</p>, <p class="purdue-home-cta-card__tag">Purdue Memorial Union </p>, <p class="purd
ue-home-cta-card__title">Places on the patio </p>, <p class="purdue-home-cta-card__content">Find the perfect spot for studyin
g (and spending time in the sun) between classes. </p>, <p class="cta-link">Learn More</p>, <p class="purdue-home-cta-card__t
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ee the Boilermakers — long known as the “Cradle of Quarterbacks” — take on other Big Ten Conference teams.  </p>, <p class="c
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reational Sports Center</p>, <p class="purdue-home-cta-card__title">GroupX</p>, <p class="purdue-home-cta-card__content">Join
a group fitness class, work out on your schedule or try something new in a state-of-the-art facility.</p>, <p class="cta-lin
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me-cta-card__title">Climbing and Bouldering</p>, <p class="purdue-home-cta-card__content">Scale a 55-foot-tall roped climbing
wall and test your skills on the bouldering wall with more than 60 horizontal feet of terrain.</p>, <p class="cta-link">Learn
More</p>, <p class="purdue-home-cta-card__tag">Purdue University in Indianapolis</p>, <p class="purdue-home-cta-card__title">
In-demand majors </p>, <p class="purdue-home-cta-card__content">Dive into future-focused fields like AI, biomedical engineeri
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t="_blank">Purdue University<br/>610 Purdue Mall<br/>West Lafayette, IN 47907</a> </p>, <p class="contact-info__phone">
<a href="tel://7654944600">765-494-4600</a> </p>, <p><em>Last modified:</em> October 14, 2025</p>]

Gets all the <p> elements with a class attribute with value 'hide':  []

Obtaining strings:  Every Giant Leap Starts with One Small Step

Obtaining strings using contents method:  ['Every Giant Leap Starts with One Small Step']

<p class="purdue-home-cta-card__cta-text">Research News</p>

Research News

['Research News']

['Meet at the union for food and retail shops, plus the Union Club Hotel.  ']

Raw text from the 4th paragraph:
================================================
Purdue Memorial Union
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Result: The program was executed successfully and output verified.